

DOES A VECTOR AUTOREGRESSION BEAT A UNIVARIATE AUTOREGRESSION AT FORECASTING INFLATION? THE APPLICATION TO THE EURO AREA AND U.S. DATA

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ABSTRACT. We ask whether adding the unemployment rate and a short-term interest rate to a model of inflation, through a vector autoregression (VAR), gives better out-of-sample inflation forecasts than a simple univariate autoregression (AR). We build a full pipeline from multivariate time series analysis: unit-root and cointegration testing, lag selection by information criteria, reduced-form VAR estimation, residual and stability checks, Granger causality, and out-of-sample forecast evaluation with the Diebold–Mariano test. We first apply it to monthly euro-area data from the European Central Bank (2000–2025). The euro-area answer is clear. Granger causality from the extra variables to inflation is not significant, the VAR has almost the same RMSE as a lag-matched AR at every horizon and is not statistically different from it in the one-step Diebold–Mariano test, and the best univariate model, a seasonal AR(12), is the most accurate at horizons from one to six months. Even the gains over a simple no-change benchmark are only small in the one-step tests. Running the same pipeline on U.S. data gives the same answer, only stronger: although Granger causality is borderline in-sample, the VAR still does not beat the univariate AR out of sample, and at long horizons it becomes much worse. In both economies the extra cross-series information is not worth its parameter cost.

1. INTRODUCTION

Forecasting inflation is one of the most policy-relevant problems in applied time series analysis, because central banks set policy looking at inflation forecasts one to twenty-four months ahead. It feels natural to think that using more information should help, in particular real-activity and monetary variables such as unemployment and interest rates. The vector autoregression (VAR) of Sims [12] is the standard multivariate tool for using this kind of cross-series information. But a long literature, starting with Atkeson and Ohanian [1], shows that simple univariate models are very hard to beat for inflation.

This report studies that tension. The research question is:

Does a VAR in inflation, unemployment and a short-term interest rate forecast inflation more accurately than a univariate AR, and does the answer depend on the economy considered?

The question is both methodological and practical. It is methodological because it is a clean test of whether cross-series information is worth the quadratic parameter cost of a VAR, and practical because good inflation forecasts matter for policy. We build the project around two real data sets, the euro area and the United States, so that we can check whether the conclusion is robust across economies. We first develop the full methodology and run it on euro-area data, and then we repeat exactly the same pipeline on U.S. data (Section 7) to see whether the verdict also holds there.

Throughout we use standard methods from multivariate time series analysis: augmented Dickey–Fuller unit-root tests, the Johansen cointegration procedure, lag selection by information criteria, reduced-form VAR estimation, portmanteau and stability diagnostics, Granger causality, and out-of-sample forecast evaluation with the Diebold–Mariano test.

Where it helps, we recall the theory behind them (stationarity and causality of VAR processes, integrated processes, cointegration and the error-correction form) in the notation of the course [10].

Our main euro-area result is negative and clear: there is *no* significant evidence that the VAR improves on the AR. Granger causality is not significant, the VAR and a lag-matched AR have almost the same RMSE at all horizons (and cannot be told apart in the one-step test), and the best univariate model, a seasonal AR(12), is the most accurate at the shorter horizons that matter most. The reason is the parameter cost: a univariate AR captures the seasonality with twelve coefficients, but this is too expensive for a three-variable VAR.

The report is organised as follows. Section 2 recalls the theory of AR and VAR processes, integrated processes and cointegration. Section 3 describes the data and Section 4 the statistical methods, together with their assumptions and test statistics. Section 5 shows the application and the diagnostics for the euro area, Section 6 the numerical results, and Section 7 the U.S. replication. Section 8 concludes.

2. PRELIMINARIES

Throughout, $Z = (Z_t)_{t \in \mathbb{Z}} \subset L^2(\mathbb{R}^d)$ denotes *white noise*: a mean-zero sequence with $\text{Cov}(Z_t) = \Sigma$ positive definite and $\text{Cov}(Z_t, Z_s) = 0$ for $t \neq s$. A process $X = (X_t)_{t \in \mathbb{Z}} \subset L^2(\mathbb{R}^d)$ is (*weakly*) *stationary* if its mean is constant and its autocovariances $\Gamma_X(h) = \text{Cov}(X_{t+h}, X_t)$ depend only on the lag h . We write L for the lag operator, $L^k X_t = X_{t-k}$.

2.1. Autoregressive processes. The univariate autoregression is just the case $d = 1$ of the following definition.

Definition 2.1 (VAR(p)). Let $p \in \mathbb{N}$ and $\Phi_1, \dots, \Phi_p \in \mathbb{R}^{d \times d}$. A stationary process X is a *vector autoregressive process of order p* if

$$(1) \quad X_t = c + \Phi_1 X_{t-1} + \dots + \Phi_p X_{t-p} + Z_t, \quad t \in \mathbb{Z},$$

or compactly $\Phi(L)X_t = c + Z_t$ with the matrix polynomial $\Phi(z) = I_d - \Phi_1 z - \dots - \Phi_p z^p$, $z \in \mathbb{C}$.

Normalising the coefficient of X_t to I_d makes Z_t the new information in X_t once the linear effect of the past is removed. In the causal stationary case, with the coefficients known, Z_t is the *one-step-ahead linear prediction error*, that is, the innovation. Existence and the moving-average form depend on a single condition.

Assumption 2.2 (Causality / stability). $\det \Phi(z) \neq 0$ for all $z \in \mathbb{C}$ with $|z| \leq 1$, i.e. all roots of $\det \Phi(z)$ lie strictly outside the closed unit disk.

Under Assumption 2.2 the model has a unique stationary and causal solution with an absolutely convergent infinite moving-average representation, so forecasts have finite error variance and revert to the mean [10]. In companion form the same assumption says that all eigenvalues of the $dp \times dp$ companion matrix lie inside the unit circle. The two forms are equivalent, because the nonzero eigenvalues of the companion matrix are the reciprocals of the roots of $\det \Phi(z)$, so “roots outside the closed unit disk” and “eigenvalues inside the unit circle” mean the same thing [10]. In practice we check stability using the eigenvalues.

Remark 2.3 (Nesting). The inflation equation of a VAR(p) contains the lags of *all* components of X_t . The univariate AR(p) of the *same order* is then the restricted model where the cross-variable lag coefficients are set to zero, so the AR(p) is *nested* in the VAR(p). This nesting needs matched lag orders: it holds for our VAR(2) against AR(2), but not for VAR(2) against the higher-order AR(12). This both motivates the comparison and tells us which forecast-accuracy test to use (Section 4).

2.2. Integrated processes and cointegration. Macroeconomic levels are usually not stationary. The basic example is the random walk $Y_t = Y_{t-1} + Z_t$, where $\text{Cov}(Y_t) = t\Sigma$ grows without bound while $\Delta Y_t = Z_t$ is stationary.

Definition 2.4 (Integrated process). Y is *integrated of order r* , written $Y \sim I(r)$, if $\Delta^r Y$ is weakly stationary and r is the smallest nonnegative integer with this property.

We work with $I(0)$ and $I(1)$ processes. If a vector X of $I(1)$ series has stationary linear combinations, then differencing every component throws them away, and this is what motivates the cointegration framework.

Definition 2.5 (Cointegration). An $I(1)$ process X is *cointegrated* of rank r if there exist r linearly independent vectors $\beta_1, \dots, \beta_r \in \mathbb{R}^d$ such that each $\beta_i^\top X_t$ is $I(0)$. The β_i are the cointegrating (long-run) relations.

Lemma 2.6 (Error-correction representation). *The VAR(p) model (1) can be written in error-correction form*

$$(2) \quad \Delta X_t = c + \Pi X_{t-1} + \sum_{j=1}^{p-1} A_j \Delta X_{t-j} + Z_t, \quad \Pi = \Phi_1 + \dots + \Phi_p - I_d, \quad A_j = -(\Phi_{j+1} + \dots + \Phi_p).$$

If $X \sim I(1)$ is cointegrated of rank r , then $\text{rank}(\Pi) = r$ and $\Pi = \alpha\beta^\top$ with $\alpha, \beta \in \mathbb{R}^{d \times r}$; if $r = 0$, then $\Pi = 0$ and a VAR in first differences is correct.

Lemma 2.6 is the basis for both the cointegration test and the modelling decision. A VAR in first differences is appropriate exactly when $\text{rank}(\Pi) = 0$, that is, when there is no cointegration. Otherwise the error-correction term ΠX_{t-1} has to be kept, which gives a vector error-correction model (VECM).

3. DATA

Euro area (primary data set). We use three monthly euro-area (reference area U2) series obtained directly from the European Central Bank (ECB) Data Portal:

- *Inflation*: HICP all-items annual rate of change, in percent (ECB code ICP.M.U2.N.000000.4.ANR);
- *Unemployment*: harmonised unemployment rate, seasonally adjusted, in percent (ECB code LFSI.M.U2.S.UNEVRT.TOTAL0.15.74.T);
- *Interest rate*: 3-month EURIBOR money-market rate, in percent (ECB code FM.M.U2.EUR.RT.MM.EURIBOR3MD.HSTA).

The common sample is monthly, from January 2000 to December 2025 ($T = 312$). After alignment there are no missing values inside the sample. We keep the data up to December 2014 (180 months) to specify and estimate the models, and we evaluate the forecasts out of sample from January 2015 on, so the evaluation period never influences the model choice. Figure 1 shows the three series; the most visible feature is the 2021–2023 inflation surge. United States (second data set). For the U.S. replication we use the matching monthly series from FRED (Federal Reserve Bank of St. Louis): the consumer price index CPIAUCSL, turned into a year-on-year rate; the civilian unemployment rate UNRATE; and the effective federal funds rate FEDFUNDS. They correspond to the euro-area variables (a headline inflation rate, a real-activity variable and a policy-related short rate), so the same methodology applies without any change. We report the U.S. results in Section 7.

Limitations of the data. The series are *revised* (final-vintage) data and not the real-time vintages that a forecaster would actually have had. This gives all models a small and equal information advantage and does not change the relative AR-versus-VAR comparison. The euro-area aggregate also has a country composition that changes over the sample.

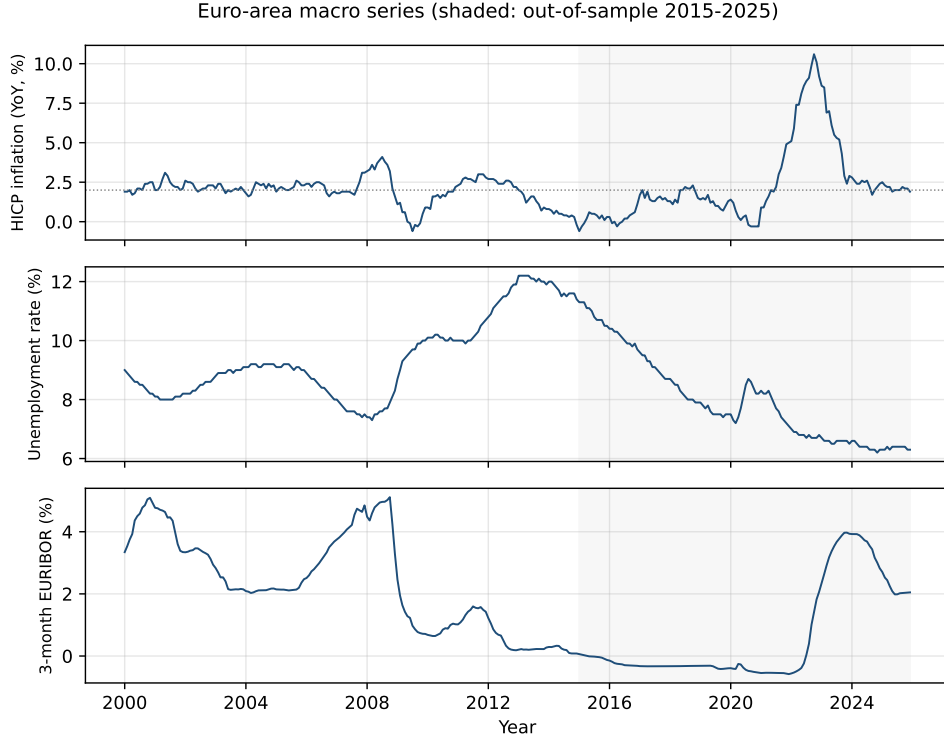


FIGURE 1. Euro-area HICP inflation, unemployment, and the 3-month EURIBOR rate, 2000–2025. The shaded region is the out-of-sample evaluation period (2015–2025).

4. METHODS

Out-of-sample accuracy is our final criterion, but the tests below are still needed: they are the justification for the modelling choices. The unit-root and cointegration tests justify modelling the system in first differences without an error-correction term, the information criteria fix the lag order, and the residual and stability diagnostics show that the fitted models satisfy the assumptions under which their forecasts, and the Diebold–Mariano comparison between them, are valid. A forecast comparison based on badly specified models would not tell us much, whatever its RMSE.

4.1. Unit-root testing. We check stationarity with the augmented Dickey–Fuller (ADF) test [4, 11]. For a chosen augmentation order k we estimate

$$(3) \quad \Delta X_t = a + \gamma X_{t-1} + \sum_{j=1}^k \psi_j \Delta X_{t-j} + Z_t,$$

and test $H_0 : \gamma = 0$ (a unit root) against $H_1 : \gamma < 0$ (stationarity), using the one-sided t -statistic $\tau = \hat{\gamma} / \text{se}(\hat{\gamma})$ and the Dickey–Fuller critical values (the ψ_j absorb short-run serial correlation, and k is chosen by AIC). The test is one-sided, so a strongly negative τ is evidence against a unit root. It is important that failing to reject H_0 is not a proof of a unit root; it only means the test did not find enough evidence against it. For this reason we phrase the outcome as “treat the level as non-stationary for modelling.”

4.2. Cointegration testing. Before differencing an $I(1)$ system we test its cointegration rank with the Johansen trace test [8], which is a likelihood-ratio test on the rank of Π in (2). With $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_d$ the ordered eigenvalues of the associated reduced-rank regression,

the trace statistic for $H_0 : \text{rank}(\Pi) \leq r$ is

$$(4) \quad \text{tr}(r) = -T \sum_{i=r+1}^d \ln(1 - \hat{\lambda}_i),$$

and its limiting distribution is non-standard and tabulated. We test in sequence starting from $r = 0$; the estimated rank is the first r that is not rejected, and rank 0 justifies a VAR in first differences (Lemma 2.6).

4.3. Lag selection. We choose lag orders with information criteria. For a VAR(p) on T observations of dimension d , with estimated innovation covariance $\hat{\Sigma}(p)$,

$$(5) \quad \begin{aligned} \text{AIC}(p) &= \ln \det \hat{\Sigma}(p) + \frac{2}{T} pd^2, & \text{BIC}(p) &= \ln \det \hat{\Sigma}(p) + \frac{\ln T}{T} pd^2, \\ \text{HQIC}(p) &= \ln \det \hat{\Sigma}(p) + \frac{2 \ln \ln T}{T} pd^2. \end{aligned}$$

BIC and HQIC are consistent for the true order, while AIC is efficient for one-step prediction but tends to overfit [10]. The penalty term pd^2 makes extra lags very costly once $d > 1$. For the univariate AR we estimate every candidate order on a common sample (holding back the maximum lag), so that the criteria can be compared across orders.

4.4. Estimation. The reduced-form VAR is estimated equation by equation by ordinary least squares. Here this coincides with the efficient generalised least squares (SUR) estimator, because every equation has the same regressors [10]. The univariate AR is estimated by conditional maximum likelihood. We use standard implementations instead of coding the estimators ourselves (Section 5).

4.5. In-sample diagnostics. We check the model assumptions of Section 2 as follows. *Stability* (Assumption 2.2) is checked through the companion-matrix eigenvalues. *Residual whiteness* is tested with a portmanteau (Ljung–Box) statistic. In the univariate case,

$$(6) \quad Q^*(h) = T(T+2) \sum_{j=1}^h \frac{\hat{\rho}_j^2}{T-j} \stackrel{a}{\sim} \chi_{h-p}^2,$$

where $\hat{\rho}_j$ are the residual autocorrelations and the degrees of freedom are reduced by the number p of fitted autoregressive coefficients. The multivariate version (Hosking) tests the joint whiteness of the whole VAR residual vector. For *Granger causality* [7], in a stable VAR(p) the group of variables $X_{(1)}$ does not Granger-cause $X_{(2)}$ if the coefficients on the lags of $X_{(1)}$ in the $X_{(2)}$ equation are jointly zero; this is an F -/Wald test that is asymptotically χ^2 . We test whether unemployment and the interest rate Granger-cause inflation.

4.6. Forecast evaluation. Forecasts are produced on an *expanding window*. At each origin the models are re-estimated on all data available up to that origin (no look-ahead), inflation is forecast h steps ahead by iterating (1), and, since the models are fitted to first differences, the level forecast is obtained by adding up the forecast changes, $\hat{\pi}_{t+h} = \pi_t + \sum_{i=1}^h \hat{\Delta}\pi_{t+i}$. Accuracy is measured by the root mean squared error (RMSE) and the mean absolute error of the level forecasts, against a no-change (random-walk) benchmark $\hat{\pi}_{t+h} = \pi_t$, which is a hard baseline for inflation [1]. Equal predictive accuracy of two forecasts is tested with the Diebold–Mariano test [5]. With squared-error loss and loss differential $d_t = e_{1,t}^2 - e_{2,t}^2$, under $H_0 : \mathbb{E}[d_t] = 0$,

$$(7) \quad \text{DM} = \frac{\bar{d}}{\sqrt{\hat{\omega}^2/n}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \hat{\omega}^2 = \hat{\gamma}_0 + 2 \sum_{k=1}^L \left(1 - \frac{k}{L+1}\right) \hat{\gamma}_k,$$

TABLE 1. Augmented Dickey–Fuller p -values on the euro-area training sample (2000–2014). A small value is evidence against the unit-root null.

Series	Level	First difference
Inflation	0.448	0.000
Unemployment	0.502	0.024
Interest rate	0.495	0.004

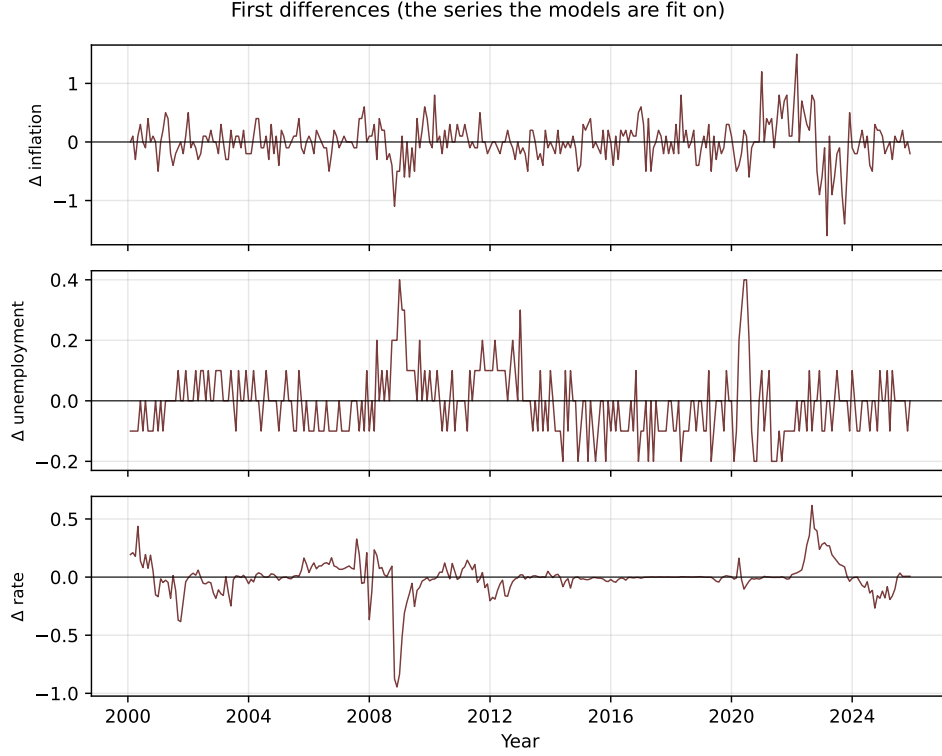


FIGURE 2. First differences of the three euro-area series; these are the series the models are fit on.

where $\hat{\omega}^2$ is a Newey–West (Bartlett) long-run variance estimate of d_t . When one model is nested in the other (the matched AR(2) and the random walk are restrictions of the VAR(2), see Remark 2.3), the Diebold–Mariano test is not exactly sized and the Clark–West correction [3] would be more precise; the comparison of VAR(2) with the non-nested AR(12) is not affected. Since the differences we find are small and not significant, this does not change our conclusions.

5. EMPIRICAL ANALYSIS (EURO AREA)

All computations are done in Python with the `statsmodels` package. The self-contained code that comes with the submission reproduces every number and figure below.

Stationarity. Table 1 reports the ADF p -values (3) on the training sample. All three series fail to reject a unit root in levels, but they reject clearly in first differences. So we treat the system as $I(1)$ and model it in first differences; Figure 2 shows the differenced series moving around a constant mean.

Cointegration. The Johansen trace statistic (4) at rank 0 is 30.76, just above the 5% critical value of 29.80, while higher ranks are not rejected. This is a *borderline* rejection that points to at most one cointegrating relation, and the evidence is weak and depends

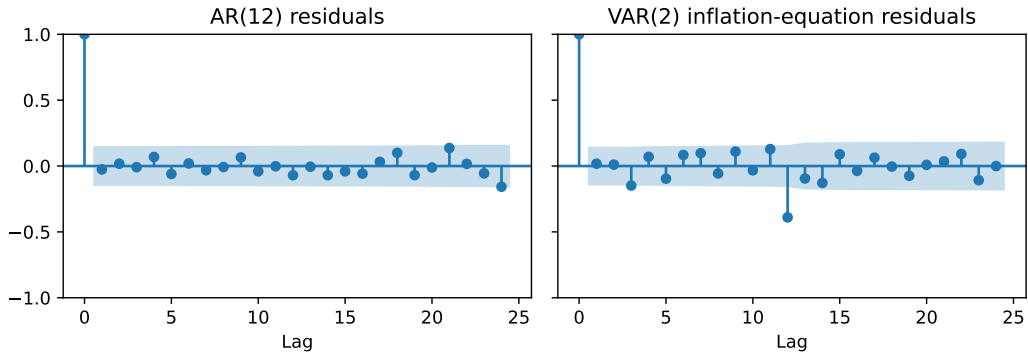


FIGURE 3. Residual autocorrelation functions. The AR(12) residuals (left) lie within the confidence band, whereas the VAR(2) inflation-equation residuals (right) still show a spike near lag 12: the parsimonious VAR cannot capture the seasonality that AR(12) absorbs.

on the sample. We keep the VAR in differences instead of moving to a VECM, for three reasons. First, the statistic is only just above the 5% value (30.76 vs. 29.80), and the Johansen test is known to reject too often in small samples, so one rank-1 relation is not really established; in empirical work such a borderline number is usually read together with the rest of the evidence and not as a final decision. Second, forcing in one cointegrating vector that we cannot estimate reliably would add error-correction parameters that may hurt rather than help the out-of-sample forecast, which is our real object. Third, the U.S. system (Section 7) shows no cointegration at all (rank 0), so a VAR in differences is the specification that fits both economies in the same way. Cointegration is therefore not the object of this study; we note the VECM (Lemma 2.6) only as a possible refinement.

Lag selection and estimation. For differenced inflation, AIC and HQIC select 12 lags while BIC selects 2. The lag-12 choice reflects the yearly seasonal pattern in monthly inflation, which a univariate AR can capture cheaply, and we take **AR(12)** as the best univariate model. For the VAR, BIC selects 1 lag and AIC/HQIC select 2; because of the pd^2 parameter cost in (5) (a VAR(12) would need over 100 slope coefficients), we take the parsimonious **VAR(2)**.

Diagnostics. The AR(12) residuals show no strong autocorrelation. The Ljung–Box statistic (6), with degrees of freedom adjusted for the twelve fitted lags, does not reject at lags 18 or 24 ($p = 0.26$ and 0.12), and the residual ACF stays inside the band (Figure 3, left). The VAR(2) is stable (all companion eigenvalues inside the unit circle), but its residuals are *not* white (Hosking $p = 0.013$), and the inflation-equation residual ACF still has a spike at lag 12 (Figure 3, right): the small VAR cannot capture the seasonality that AR(12) absorbs. Granger causality from unemployment and the interest rate to inflation is not significant ($p = 0.456$), which is already a sign that the cross-series information adds little.

We treat seasonality on purpose through the lag structure and not by pre-adjusting the data, so that the series enter raw and both models see the same information and the comparison stays clean. Monthly inflation has a yearly cycle, and to capture it a model needs a lag reaching twelve months back. A univariate AR pays for this with twelve scalar coefficients, which is why AR(12) whitens its residuals. The same reach in a three-variable VAR would cost 12×3^2 slope parameters, so the VAR(2) that we can afford has to leave the lag-12 residual spike of Figure 3. This asymmetry, seasonality being cheap for a univariate model but expensive for a multivariate one, is itself part of why the VAR does not pay off. A seasonally adjusted or explicitly seasonal specification is a possible alternative, but we do not use it, because it would change the object of the comparison.

TABLE 2. Out-of-sample RMSE of one- to twelve-month-ahead euro-area inflation forecasts (2015–2025); lower is better, best in each row in bold.

h (months)	VAR(2)	AR(2)	AR(12)	No-change
1	0.399	0.402	0.369	0.425
3	0.842	0.862	0.820	0.932
6	1.479	1.505	1.445	1.595
12	2.728	2.729	2.876	2.728

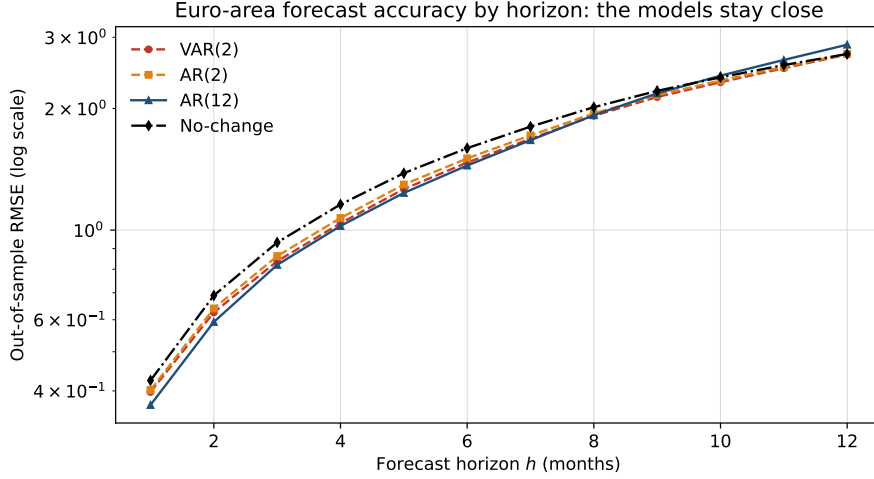


FIGURE 4. Euro-area out-of-sample RMSE by forecast horizon (log scale). All four models rise together and stay close: VAR(2) is indistinguishable from the matched AR(2), and the seasonal AR(12) is best at short horizons but crosses to become the worst by $h = 12$. Contrast the U.S. divergence in Figure 9.

6. RESULTS AND DISCUSSION

Table 2 reports the out-of-sample RMSE (2015–2025) for the VAR(2), a lag-matched AR(2), the seasonal AR(12), and the no-change benchmark, and Figure 4 plots the same errors against the forecast horizon. Three things stand out.

First, *the VAR does not beat the matched AR*. The VAR(2) and AR(2) have almost the same RMSE at every horizon, and the one-step Diebold–Mariano test (7) does not reject equal accuracy ($p = 0.42$). Adding unemployment and the interest rate to inflation’s own past changes the forecast almost not at all, exactly as the insignificant Granger test suggests.

Second, *the best univariate model beats the VAR*. The seasonal AR(12) has the lowest RMSE at $h = 1, 3, 6$, and the VAR does not improve on it (one-step Diebold–Mariano $p = 0.16$). The reason is again the parameter cost made explicit in (5): capturing inflation’s seasonality needs about twelve lags, which costs a univariate AR twelve coefficients but a three-variable VAR more than a hundred. A higher-lag VAR would be even more parameter-hungry, so the parsimonious VAR(2) leaves seasonal residual autocorrelation (Figure 3), while the univariate AR(12) absorbs it. At the longest horizon ($h = 12$) the picture turns around for a different reason: the many seasonal lags of AR(12) accumulate error and it becomes the *worst* model, while VAR and no-change end up almost tied. Figure 5 shows that the VAR(2) and AR(12) forecast paths are almost impossible to tell

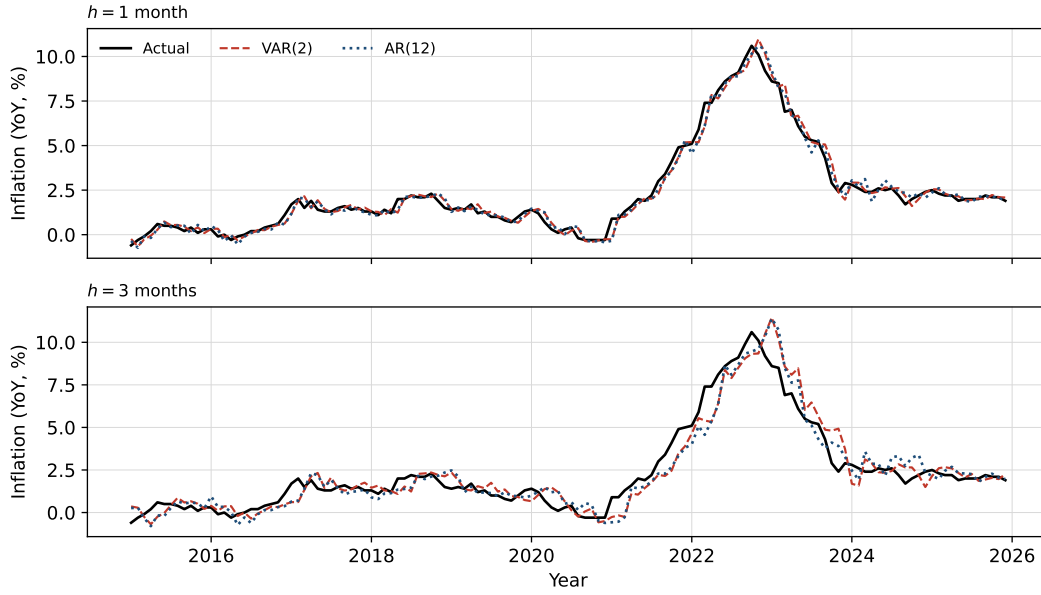
Euro-area inflation forecasts at horizons $h = 1, 3$ (2015–2025)

FIGURE 5. Euro-area inflation forecasts at horizons $h = 1, 3$, 2015–2025. VAR(2) and AR(12) stay nearly identical at both horizons and both lag the sharp 2021–2022 surge, increasingly so as the horizon grows.

apart at $h = 1, 3$; both follow inflation closely and both lag the 2021–2022 surge, more and more as h grows.

Because the level forecasts lie so close to each other, Figure 6 instead plots the one-step forecast *errors* directly. Panel (a) shows that the VAR(2), AR(12) and no-change error paths are almost the same, and that all three miss the 2021–2022 surge together. Panel (b) adds up the squared-error difference against AR(12): the curve stays close to zero until 2021 and then drifts up a little, so AR(12) builds a small advantage over both VAR(2) and the no-change benchmark, driven by the surge and not systematic. This is the same near-tie that the RMSE table and the one-step Diebold–Mariano test give in numbers.

Third, *even beating the no-change benchmark is hard*. In the one-step comparison the improvements of the VAR and AR(12) over the random walk are only small (Diebold–Mariano $p = 0.075$ and 0.067 , borderline at 10% but not significant at 5%); we do not test the longer horizons formally.

Robustness. The conclusion does not depend on the lag order. If we re-run the comparison with $p = 1$ and $p = 2$ for both the VAR and the matched AR (Table 3), the VAR and AR stay within a few percent of each other at every horizon, and the tiny gap even changes sign. The parsimonious $p = 2$ is always a little better than $p = 1$, but the AR-versus-VAR verdict does not change. We also checked that the qualitative conclusion is stable to the choice of training/evaluation split.

Relation to the literature. This finding fits well with the inflation-forecasting literature: simple univariate models that absorb the slow trend and the seasonality are very hard to beat, and richer multivariate models often fail to pay their parameter cost out of sample [1, 13, 6]. Our euro-area results reproduce this pattern in a small and transparent system.

7. APPLICATION TO U.S. DATA

The second data set is the United States. The U.S. series match the euro-area variables (Section 3) and are also monthly, so the *whole* methodology of Sections 4–6 applies without

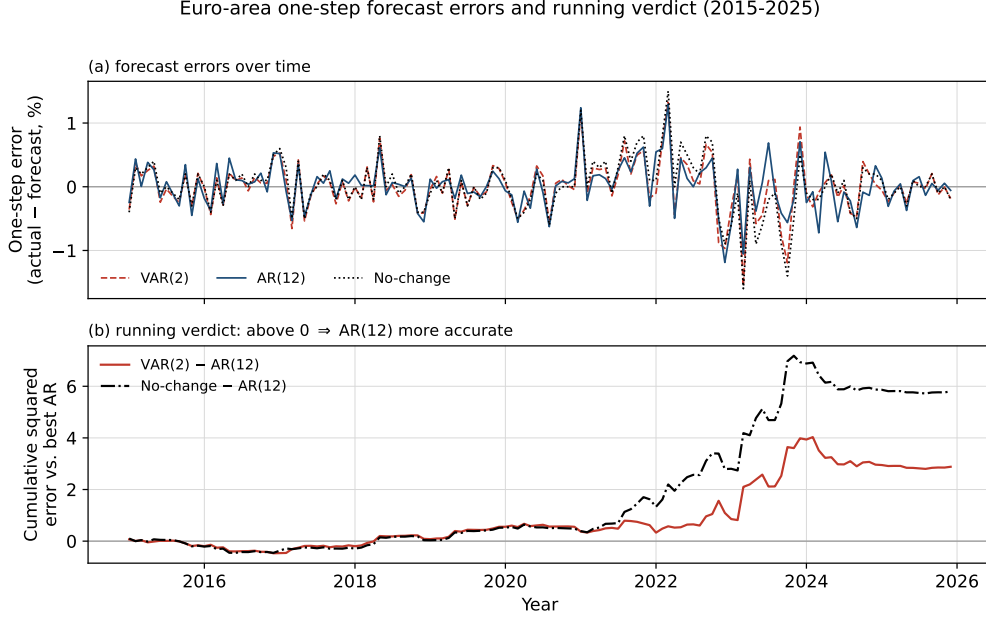


FIGURE 6. Euro-area one-step forecast errors (a) and the cumulative squared-error difference against AR(12) (b), 2015–2025. In (b) a rising line means AR(12) is the more accurate model; the small, late advantage confirms that no specification systematically dominates.

TABLE 3. Robustness to the lag order: out-of-sample euro-area RMSE for $p = 1$ versus $p = 2$.

h	VAR(1)	VAR(2)	AR(1)	AR(2)
1	0.410	0.399	0.408	0.402
3	0.914	0.842	0.904	0.862
6	1.610	1.479	1.575	1.505
12	2.828	2.728	2.768	2.729

change; the same code runs on either data set when we swap only the data-loading step, and we keep the same 2000–2014 training and 2015–2025 evaluation split. The question is whether the AR-versus-VAR verdict also holds in a second economy with a different monetary regime. Figure 7 shows the series; note the sharp 2020 unemployment spike, which will matter later.

Stationarity and cointegration. The ADF p -values (Table 4) treat inflation and unemployment as $I(1)$, while the funds rate is borderline stationary in levels ($p = 0.017$). In first differences inflation and the rate reject clearly, but differenced unemployment rejects only at 10% ($p = 0.086$), which is a hint of near- $I(2)$ behaviour caused by the 2020 spike. For comparability we model all three in first differences, as before. The Johansen trace statistic (4) at rank 0 is 25.70, *below* the 5% critical value 29.80. So, unlike the borderline euro-area case, here there is no cointegration (rank $\Pi = 0$), and a VAR in differences is clearly the right choice (Lemma 2.6).

Lag selection and diagnostics. For differenced inflation the criteria disagree: AIC picks 24 (the boundary of the search, which is a sign of overfitting), BIC picks 12, and HQIC picks 15. We take the richest one, **AR(24)**, to keep the comparison favourable to the univariate side. For the VAR, AIC and FPE pick 12 while BIC picks 1 and HQIC picks 2, so we carry *both* the parsimonious **VAR(2)** and the AIC/FPE-optimal **VAR(12)** forward.

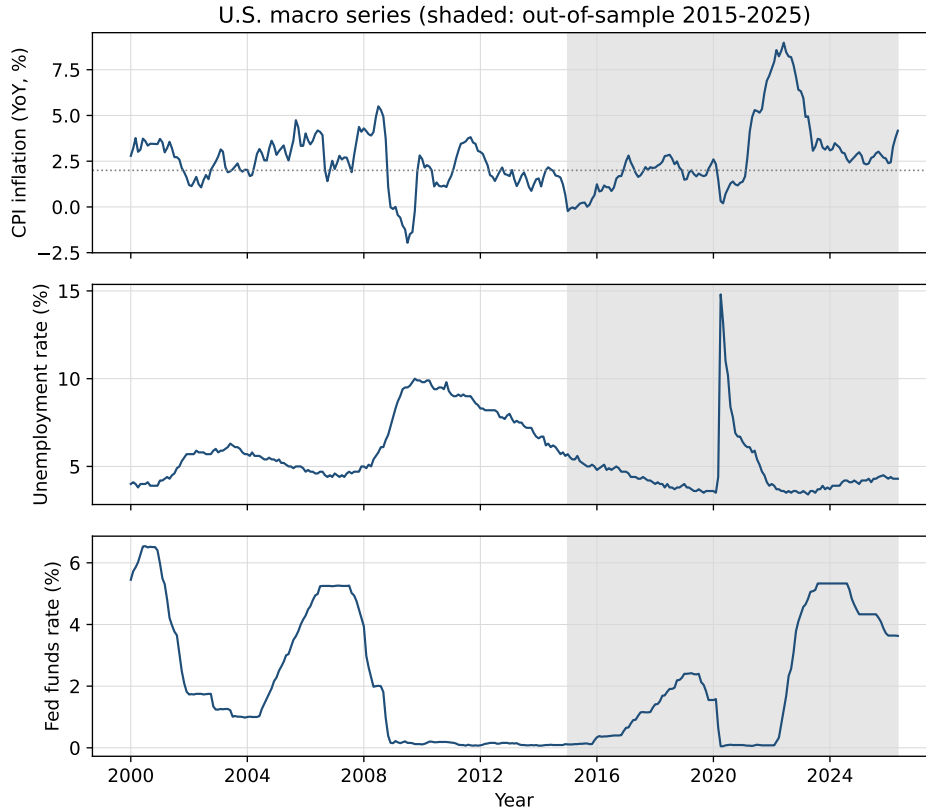


FIGURE 7. U.S. CPI inflation, unemployment, and the effective federal funds rate, 2000–2025. The shaded region is the out-of-sample evaluation period; the 2020 unemployment spike is the COVID shock that later destabilises the VAR forecasts.

TABLE 4. Augmented Dickey–Fuller p -values on the U.S. training sample (2000–2014). A small value is evidence against the unit-root null.

Series	Level	First difference
Inflation	0.260	0.001
Unemployment	0.229	0.086
Interest rate	0.017	0.011

Both VARs are stable, but neither has white residuals (Hosking $p = 0.000$ and 0.001 ; Figure 8), and even AR(24) keeps some autocorrelation (Ljung–Box $p = 0.002$ at lag 30). Unlike the euro area, Granger causality from unemployment and the rate to inflation is now *borderline significant* ($p = 0.051$ for VAR(2), 0.013 for VAR(12)). So in-sample the extra variables look informative, which makes the out-of-sample results below even more surprising.

Forecast results. The out-of-sample RMSE (Table 5) is the euro-area conclusion in a sharper form: at every horizon the VARs are *worse* than AR(24), and at long horizons they do not just lose, they *explode*. Figure 9 plots RMSE against the horizon on a log scale: AR(24) and the no-change benchmark rise slowly and stay together, while both VARs grow roughly geometrically and reach RMSE 14.5 (VAR(2)) and 26.9 (VAR(12)) at $h = 12$, against 2.5 and 2.3. The reason is the 2020 unemployment spike (Figure 7): in a differenced VAR this huge cross-variable shock enters the inflation equation and, iterated

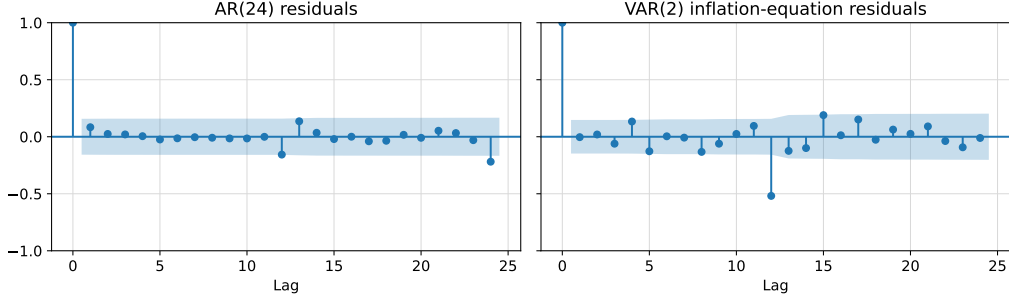


FIGURE 8. U.S. residual autocorrelation functions: AR(24) (left) and the VAR(2) inflation equation (right). Both leave some structure, consistent with the whiteness tests.

TABLE 5. Out-of-sample RMSE of one- to twelve-month-ahead U.S. inflation forecasts (2015–2025); lower is better, best in each row in bold. The VARs diverge at long horizons.

h (months)	VAR(2)	VAR(12)	AR(24)	No-change
1	0.402	0.668	0.302	0.385
3	1.215	1.693	0.800	0.881
6	3.093	5.313	1.392	1.375
12	14.467	26.854	2.502	2.270

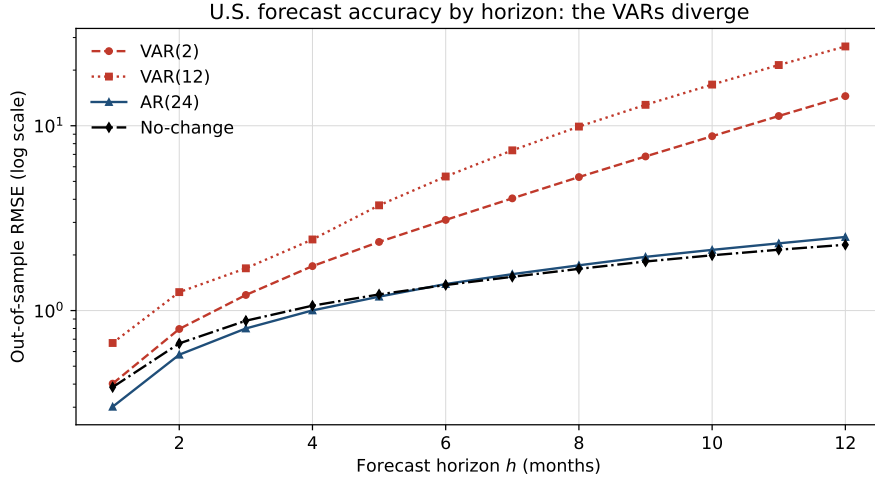


FIGURE 9. U.S. out-of-sample RMSE by forecast horizon (log scale). AR(24) and the no-change benchmark track together; both VARs diverge roughly geometrically, the AIC/FPE-optimal VAR(12) fastest of all.

forward, builds up into very large multi-step errors. It is worth noting that the AIC/FPE-preferred VAR(12) is the worst model of all, which is exactly the overfitting that the pd^2 penalty in (5) warns about. Figure 10 shows this directly: at $h = 1$ all models follow inflation (though VAR(12) already wobbles at the 2020 shock), but by $h = 3$ both VARs drop to about -8% around 2020 while AR(24) stays close; this is the seed of the long-horizon divergence in Figure 9.

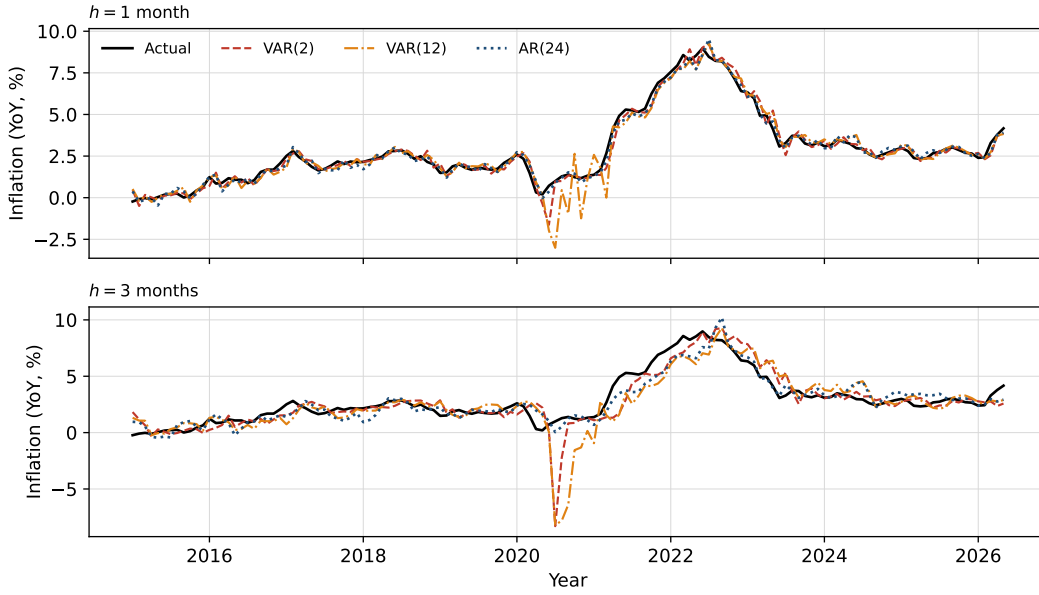
U.S. inflation forecasts at horizons $h = 1, 3$ (2015–2025)

FIGURE 10. U.S. inflation forecasts at horizons $h = 1, 3$, 2015–2025. At $h = 1$ all models track inflation (VAR(12) already wobbles at the 2020 shock); by $h = 3$ both VARs plunge to about -8% around 2020 while AR(24) stays close to actual. This is the instability that, iterated further, produces the long-horizon RMSE explosion of Table 5 and Figure 9.

The Diebold–Mariano tests (Table 6) have to be read carefully. At $h = 1$ no pair is distinguishable at 5%: VAR against AR(24) gives $p = 0.113$ and 0.133 , and VAR(2) against no-change gives $p = 0.705$. This is *not* evidence of a close contest. The one-step RMSE gaps are real (AR(24) 0.302 against VAR(2) 0.402), but the VAR loss differentials are heavy-tailed because of the unstable episodes, and this inflates the long-run variance in (7) and stops the test from rejecting. As in the euro area, only AR(24) against no-change comes close to significance ($p = 0.071$). We keep these statements to the one-step horizon, the only one we test, and we note that the Clark–West caveat (Remark 2.3) applies only to the nested no-change benchmark and not to the non-nested AR(24).

Figure 11 makes this visible. Panel (a) shows the VAR(2) one-step errors spiking at the 2020 shock (the $+2.3$ point), where the AR is comparatively under control. Panel (b) adds up the squared-error difference against AR(24): the curve is almost flat until 2020 and then jumps in two steps (the 2020 shock and the 2021–2022 surge) as AR(24) pulls clearly ahead of *both* the VAR and the no-change benchmark. This shows that the VAR’s losses are concentrated in a few episodes, which is exactly why the one-step Diebold–Mariano test, which averages a heavy-tailed loss differential, does not reject even though there is a clear RMSE gap.

Cross-region verdict. The verdict travels, and in the U.S. it is even stronger: while the euro-area VAR was only no better than the AR, the U.S. VAR actually hurts, and badly so at long horizons. What is important is that the borderline in-sample Granger significance gives *no* out-of-sample gain, which is a reminder that in-sample predictability and out-of-sample accuracy are two different things. In both economies, and under two different monetary regimes, adding unemployment and an interest rate to a univariate model of inflation does not improve the forecast, and can even make it much worse.

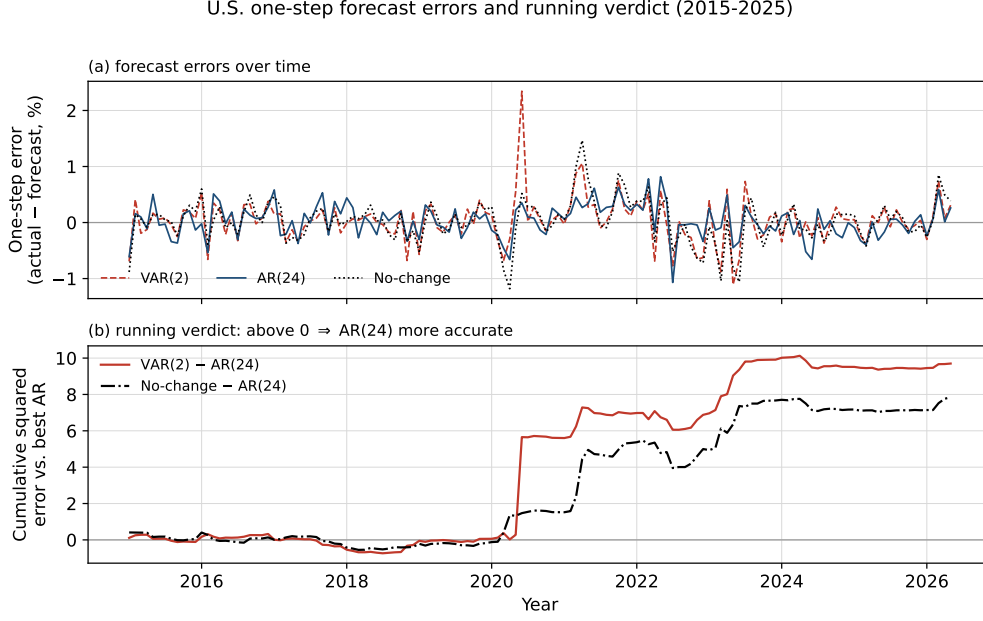


FIGURE 11. U.S. one-step forecast errors (a) and the cumulative squared-error difference against AR(24) (b), 2015–2025. In (b) a rising line means AR(24) is the more accurate model; the two upward steps at 2020 and 2021–2022 show the VAR’s losses are concentrated in those episodes.

TABLE 6. One-step Diebold–Mariano p -values (7) for U.S. inflation forecasts (equal squared-error accuracy). No pair is distinguishable at 5%.

Comparison	p -value
VAR(2) vs. AR(24)	0.113
VAR(12) vs. AR(24)	0.133
VAR(2) vs. no-change	0.705
VAR(12) vs. no-change	0.193
AR(24) vs. no-change	0.071

8. CONCLUSION AND OUTLOOK

For euro-area inflation over 2000–2025 we find *no significant evidence that a VAR in inflation, unemployment and a short-term interest rate improves on a univariate AR*. Granger causality is not significant, the VAR matches a lag-matched AR in RMSE at all horizons (and cannot be told apart from it in the one-step test), and the best univariate model, a seasonal AR(12), is the most accurate at the one-to-six-month horizons that matter most. The main reason is the quadratic parameter cost of the VAR: the seasonality that a univariate AR captures with twelve coefficients is too expensive for a three-variable VAR, so the cross-series information adds estimation noise instead of signal.

The study has clear limitations. The cointegration test is borderline, so a VECM (Lemma 2.6) might behave differently at long horizons. We use final-vintage data, a single training/evaluation split, and one-step significance tests. The evaluation window also covers the COVID-19 shock of 2020 and the 2021–2023 inflation surge, an unusual regime shift that raises the absolute error levels and puts the constant-parameter assumption of both models under stress. Since this shock hits all candidate models in a similar way, we expect the *relative* AR-versus-VAR ranking to be robust to it, but the error sizes and any

single-window result should still be read with this episode in mind. Natural extensions are a Clark–West evaluation for the nested comparison and multi-horizon Diebold–Mariano tests.

On a more basic level, the AR and the VAR share the same *linear, fixed-coefficient* structure. The 2021–2023 surge, which every model here missed, suggests that this structure, rather than the univariate-versus-multivariate choice, may be the real limit on accuracy. Model families that relax it are therefore a natural direction for future work: time-varying-parameter and regime-switching state-space models in the spirit of the Kalman filter, and nonlinear machine-learning methods, both try to capture the nonlinear, regime-dependent behaviour of inflation. We add the caveat that the evidence on whether such richer models reliably beat simple univariate benchmarks for inflation is still mixed [6], so this is an open question and not a settled improvement.

The U.S. replication (Section 7) confirms that this conclusion, that more variables do not automatically mean better inflation forecasts, also holds in a second economy, and there it is even stronger: the U.S. VARs not only fail to beat the univariate AR but diverge strongly at long horizons, and the borderline in-sample Granger causality gives no out-of-sample gain. The fact that the same verdict shows up in two economies with different monetary regimes is the main empirical result of this study.

CODE AVAILABILITY

The complete, self-contained code that reproduces every number and figure in this report is publicly available at https://github.com/ZhukovAndrii/MTSA_project. The two Jupyter notebooks download the data directly from the ECB Data Portal and FRED and re-create all reported results and figures.

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